OLYMPIC SPORTS PERFORMANCE DASHBOARD

Data Source: Kaggle | 120 Years of Olympic History

Analyzing 120 Years of Olympic History (1896–2016) using Microsoft Power BI

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Business Intelligence & Data Visualization

PROJECT OVERVIEW & OBJECTIVES

BACKGROUND

Olympic data captures athlete participation and performance across eras. Analysing it reveals trends in participation and success factors.

PROJECT OBJECTIVE

Create an interactive Power BI dashboard to explore participation, athlete profiles, and medal performance over time.

PROJECT SCOPE

Period: 1896–2016
Source: Kaggle
Tool: Power BI
Focus: Summer & Winter

BUSINESS PROBLEM & QUESTIONS

KEY QUESTIONS

- Which countries and sports dominate medal counts?
- How has participation and performance evolved over time?
- What are the physical characteristics of successful athletes?
- How do Summer and Winter Olympics differ?
- Which age groups achieve the highest Olympic success?

PROBLEM STATEMENT

National Olympic Committees and sport governing bodies need a clearer, data-driven view of the factors behind Olympic success to spot trends, allocate resources, and develop athletes strategically.

DATASET & DATA PREPARATION

DATASET OVERVIEW

Name: 120 Years of Olympic History
Period: 1896 – 2016
Records: 135,571 Athlete Entries

KEY VARIABLES

Athlete Name | Age | Height | Weight | Gender | Country | Sport | Event | Medal | Year

POWER QUERY TRANSFORMATIONS

- Created BMI Column
- Defined Age Groups
- Categorized Performance
- Defined Olympic Eras
- Created Reference Queries

DATA MODELING: STAR SCHEMA

ARCHITECTURE

A Star Schema was implemented to optimize query performance and reporting, enabling efficient retrieval across dimensions.

FACT TABLE

Factathlete_events: Central table with measurable event data, athlete performance and medal outcomes.

KEY RELATIONSHIPS

One-to-many relationships link dimensions to the fact table for slicing by year, sport and geography.

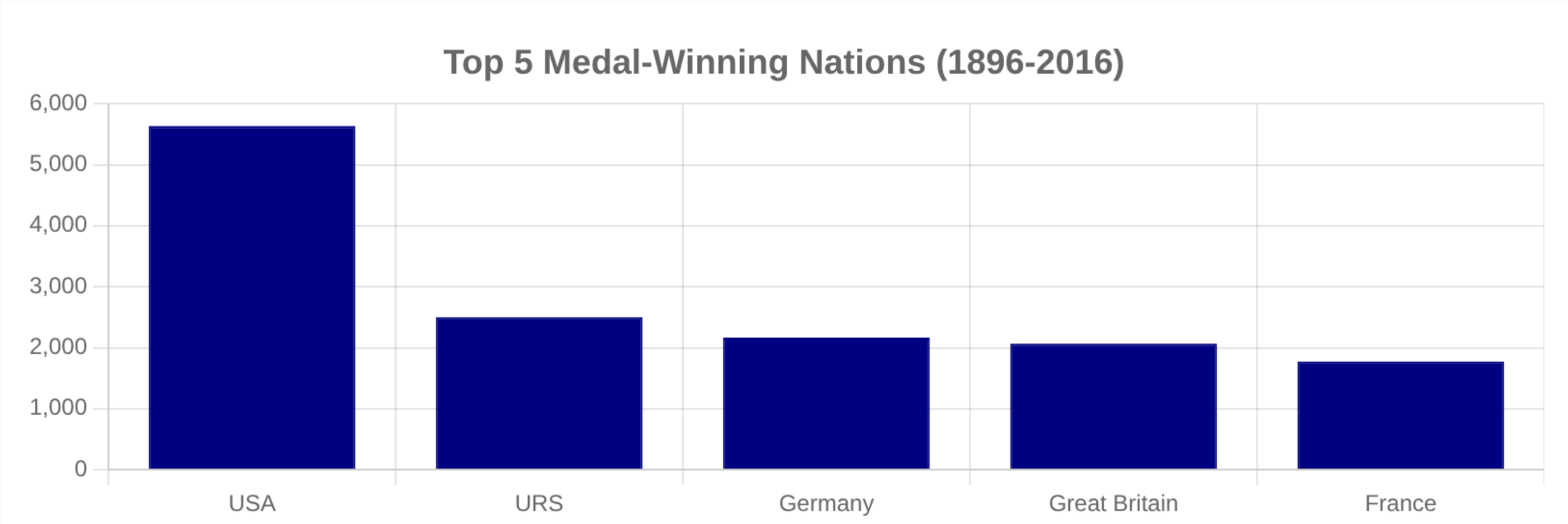
DIMENSION TABLES

DimAthlete

DimSport

Descriptive attributes for comprehensive analysis.

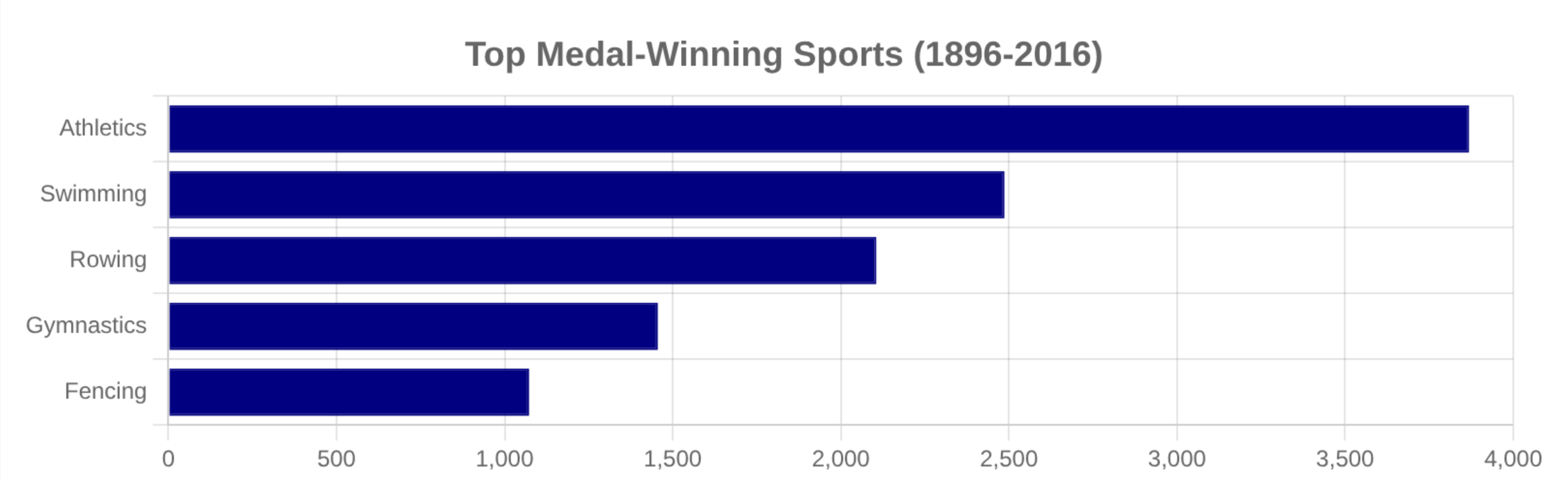
GLOBAL MEDAL DISTRIBUTION



KEY INSIGHT

The **United States** is the most successful Olympic nation with approximately **5.6K medals**, significantly outperforming all other nations.

DOMINANT OLYMPIC SPORTS



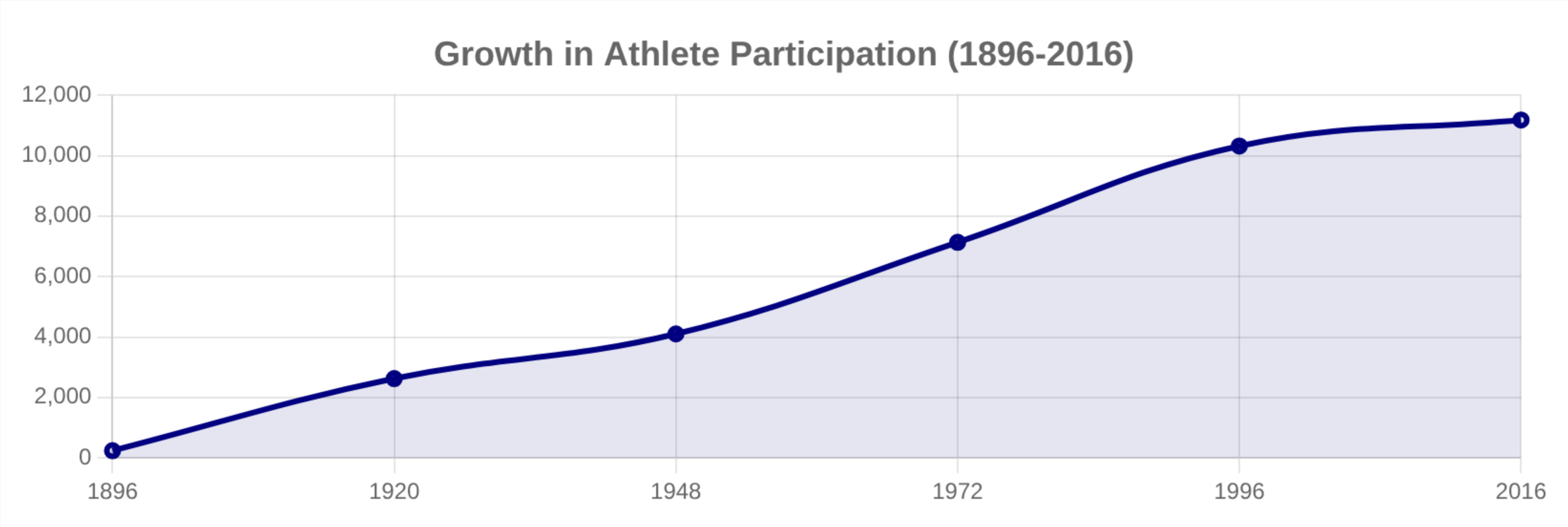
STRATEGIC INSIGHT

Athletics, **Swimming**, and **Rowing** contribute the highest share of total Olympic medals.

These "Big Three" sports represent key opportunities for medal acquisition and national ranking improvement.

Recommendation: Prioritize resource allocation and elite coaching in these high-yield disciplines.

GLOBAL EXPANSION TRENDS

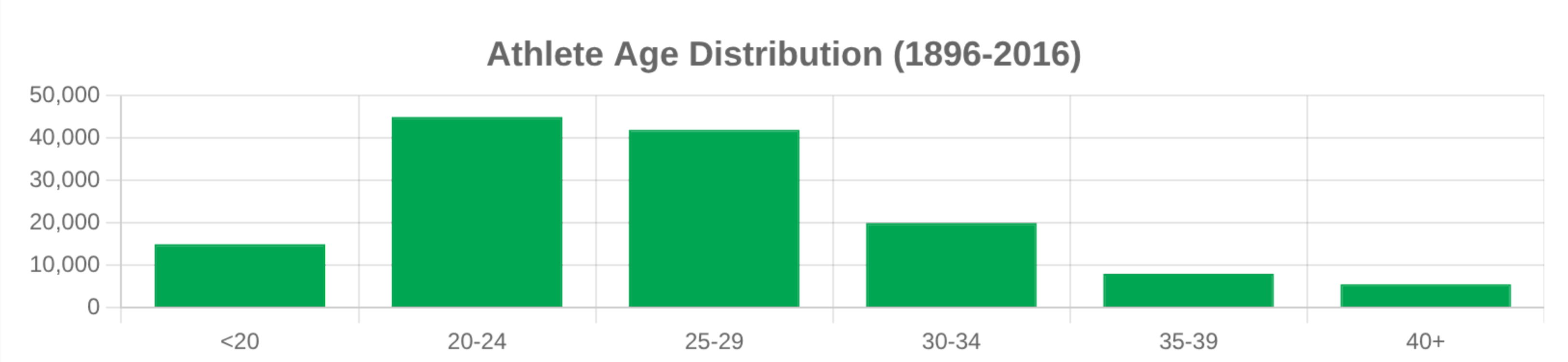


GROWTH ANALYSIS

Athlete participation and medal counts have increased substantially over time, demonstrating global expansion.

Key Drivers: Inclusion of more events, expansion of Winter Games, and increased global reach.

ATHLETE PROFILE & PEAK AGE



Athletes aged **20–29** account for the vast majority of Olympic participants, suggesting this is the window for peak physical performance.

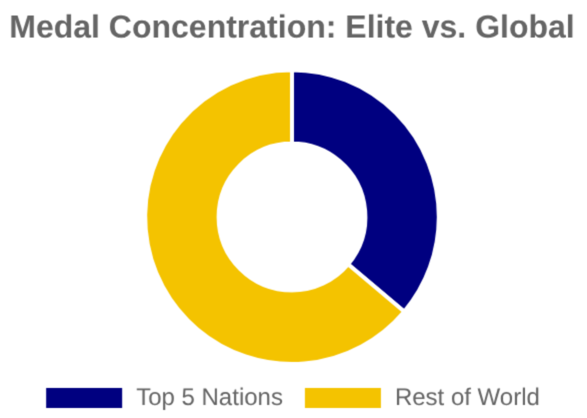
24.45
AVERAGE ATHLETE AGE

22.8
AVERAGE BMI

175cm
AVERAGE HEIGHT

70kg
AVERAGE WEIGHT

GEOGRAPHIC CONCENTRATION



KEY FINDING

Olympic success is concentrated among a relatively small number of countries with established sports infrastructure.

Consistently strong performers include:

- **United States**
- **France**
- **Great Britain**
- **Germany**
- **Italy**

Implication: Success is tied to long-term national investment and systemic development programs.

STRATEGIC RECOMMENDATIONS

RESOURCE ALLOCATION

Increase investment in high-yield sports like Athletics, Swimming, and Rowing to maximize medal potential.

YOUTH DEVELOPMENT

Strengthen programs targeting the 20-29 peak performance window to ensure a pipeline of elite talent.

BENCHMARKING

Adopt data-driven systems to monitor performance against leading nations like the USA and Great Britain.

PERFORMANCE SYSTEMS

Implement advanced monitoring systems for real-time athlete development and strategic decision-making.

TALENT IDENTIFICATION

Use physical characteristics (BMI, age, height, weight) as key metrics for identifying and nurturing potential Olympic contenders.

"Data-driven selection reduces bias and increases the probability of long-term success."

CONCLUSION & FUTURE OUTLOOK

DATA-DRIVEN FOUNDATION

The Power BI dashboard identified key factors influencing Olympic success, providing a foundation for strategic sports management.

KEY TAKEAWAYS

Analysis revealed dominant countries, high-performing sports, participation trends, and critical athlete characteristics.

STRATEGIC IMPACT

These insights enable National Olympic Committees to move beyond intuition, allowing precise resource allocation and athlete development.

LEVERAGING DATA IS NO LONGER OPTIONAL FOR OLYMPIC SUCCESS.

The future of sports performance lies in integrating business intelligence and athletic development.

Thank You!